Food Expiry Detection and Reminder System: An Integrated Barcode, OCR, and Computer Vision-Based Approach

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***Abstract*—Food waste remains a critical global issue, largely driven by consumers' inability to track the freshness of stored food items. While packaged products display printed expiry dates, fresh produce such as vegetables and fruits carry no such labeling, often resulting in premature disposal or consumption of spoiled food. This paper presents the design of an integrated Food Expiry Detection and Reminder System that unifies expiry tracking for both packaged and fresh food items within a single mobile application. For packaged products, the system identifies items through barcode scanning and extracts manufacturing and expiry dates using an on-device Optical Character Recognition (OCR) pipeline. For fresh produce, where printed dates are unavailable, a lightweight convolutional neural network performs a live visual scan of the item to classify its freshness stage, which is combined with user-specified storage conditions to estimate the remaining shelf life. All items are logged into a centralized inventory that issues configurable notifications before expiry. The proposed system is positioned as a low-cost, largely offline-capable alternative to existing solutions, most of which address either packaged-item tracking or fresh-produce freshness estimation individually, but rarely combine both within a single integrated pipeline.**

***Keywords*—food expiry detection; optical character recognition; barcode scanning; convolutional neural network; freshness estimation; food waste reduction**

I. INTRODUCTION

Food waste is widely recognized as one of the most persistent inefficiencies in modern food systems, and a significant portion of it originates not from scarcity but from poor tracking of how long stored food remains safe to consume. Households routinely discard items that are still edible simply because the purchase date has been forgotten, while, less visibly, some food is consumed past the point of safety because no clear expiry cue was available. Both outcomes point to the same underlying gap: there is no convenient, low-effort way for an ordinary consumer to keep track of the freshness of everything stored in a kitchen.

For packaged food products, this gap is partially addressed by the printed expiry date itself, provided the consumer notices and remembers it. Several existing mobile applications already automate this step by scanning the label and extracting the date using Optical Character Recognition (OCR). However, this approach is fundamentally limited to packaged goods. Fresh produce such as tomatoes, onions, bananas, and leafy vegetables carries no printed date at all, and the freshness of such items instead depends on visual cues — color, firmness, surface texture, and the presence of spots or wilting — that change gradually over time. Most existing food-tracking systems either ignore this category entirely or fall back on a static, category-based shelf-life estimate that does not reflect the actual condition of the item or how it has been stored.

This paper proposes a Food Expiry Detection and Reminder System that addresses both categories within a single mobile application. The system combines three complementary techniques: barcode-based product identification, OCR-based expiry date extraction for packaged goods, and a lightweight image-classification model that performs a live visual scan of fresh produce to estimate its freshness stage. The freshness classification is further combined with a user-specified storage condition (refrigerated, room temperature, or frozen) to adjust the predicted remaining shelf life. All tracked items are stored in a centralized inventory that issues configurable reminders before the predicted expiry date.

The principal contributions of this work are as follows: (i) a unified pipeline that treats packaged and fresh food items as first-class categories within the same system, rather than treating fresh produce as a secondary feature; (ii) a storage-condition-aware adjustment to shelf-life prediction, which existing OCR-centric systems do not provide; and (iii) a system design that favors on-device processing over continuous cloud dependency, reducing both operating cost and reliance on network connectivity.

The remainder of this paper is organized as follows. Section II reviews related work on OCR-based expiry date recognition and deep-learning-based freshness classification. Section III describes the proposed system. Section IV presents the system architecture. Section V discusses implementation details. Section VI discusses expected outcomes relative to benchmarks reported in the literature. Section VII concludes the paper and outlines directions for future work.

II. LITERATURE SURVEY

Dospinescu et al. [1] developed a mobile application that combines on-cloud image classification with OCR to track food expiry, using Firebase ML Kit for text recognition and a category-based fallback rule for items without a detectable printed date. The system was evaluated on 59 supermarket products across 12 categories and reported a product identification accuracy of 72.9% and an OCR date-extraction accuracy of 83.3% on packaged items, with the category-based fallback providing full coverage for the remaining items. The study is closest in scope to the present work, since it is one of the few systems that attempts to address both packaged and

unlabeled items within one application; however, its fresh-item handling relies entirely on static category priors rather than on the visual condition of the item itself, and the evaluation was limited to a single store and a single test device.

A substantial body of work focuses specifically on improving OCR accuracy for printed expiry dates on packaging. Seker and Ahn [3] proposed a generalized framework that combines a fully convolutional detection stage with a neural date-parsing stage capable of distinguishing thirteen different date formats, reporting a recognition accuracy of 97.74% on their purpose-built ExpDate dataset. Gong et al. [4] proposed a unified deep neural network pipeline for use-by date recognition on retail packaging, combining text detection and recognition stages into a single trainable system. More recently, Peng et al. [2] presented a framework aimed specifically at accessibility for visually impaired users, applying convolutional neural networks to both detect the expiry date region and recognize its characters. Collectively, these studies demonstrate that purpose-built detection-then-recognition pipelines substantially outperform generic, off-the-shelf OCR when applied to the specific problem of printed expiry dates, which vary widely in font, embossing, and placement.

On the fresh-produce side, Mukhiddinov et al. [5] proposed an improved YOLOv4-based model to classify fruits and vegetables as fresh or rotten, trained on a custom dataset of 12,000 images spanning five fruit and five vegetable types. Their system achieved a mean average precision of 73.5% at an IoU threshold of 0.5 on the augmented dataset, with confusion-matrix accuracy exceeding 97% for both fruit and vegetable freshness classification, and was deployed in a client-server architecture for real-time use. This body of work confirms that visual, deep-learning-based freshness classification is both feasible and reasonably accurate for common produce types, but it does not address packaged-item tracking, nor does it adjust its freshness estimate based on how or where the item is stored.

Taken together, the literature reveals a consistent division: OCR-centric systems [1]–[4] handle packaged goods well but treat fresh produce as, at best, a static fallback, while vision-based freshness classifiers [5] handle fresh produce well but do not address packaged-item date extraction or shelf-life personalization. The system proposed in this paper is designed to close this gap by combining both approaches into a single pipeline and by introducing a storage-condition-aware adjustment layer that, to the best of our knowledge, is not present in any of the systems reviewed above.

III. PROPOSED SYSTEM

A. System Overview

The proposed system is a mobile application through which a user scans or photographs a food item, after which the system routes the item through one of two pipelines depending on whether it is a packaged product or an item of fresh produce. Both pipelines converge on a shared inventory and notification module. Figure 1 (described in Section IV) summarizes this flow.

B. Barcode-Based Product Identification

For packaged products, the mobile camera is used to scan the item's barcode. The decoded identifier is matched against an open product database to retrieve the product name, category, and, where available, standard shelf-life information. This avoids the need to build and maintain a proprietary product catalog.

C. OCR-Based Expiry Date Extraction

Following product identification, an on-device OCR engine scans the label image to locate and extract the printed manufacturing and expiry dates. Because such dates appear in a wide variety of formats, the extracted text is passed through a regex-based normalization step that reconciles this variation into a single internal date representation. Where OCR confidence is low — due to glare, curvature, or low contrast — the user is prompted to confirm or manually enter the date rather than having the system silently accept an unreliable reading.

D. Live Visual Freshness Estimation for Fresh Produce

For fresh produce, where no printed date exists, the user instead points the camera at the item for a live scan. A lightweight convolutional neural network, obtained through transfer learning on a pretrained backbone, classifies the item into one of several freshness stages (for example, fresh, early ripening, near spoilage, or spoiled) based on visual indicators such as color, surface texture, and the presence of spots or wilting. The initial prototype targets a limited set of common items — banana, tomato, leafy greens, apple, onion, and potato — to keep the classification task tractable while covering items that appear most frequently in household inventories.

E. Storage-Condition-Aware Shelf-Life Adjustment

The freshness stage returned by the visual classifier is combined with a user-specified storage condition — refrigerated, room temperature, or frozen — to compute an estimated number of remaining days before the item is likely to spoil. This adjustment is applied through a condition-specific multiplier rather than a single fixed duration per food type, so that the same item produces a different estimate depending on where it is stored. Users may manually override this estimate at any time, and the system treats such overrides as a signal that can be used to refine future predictions for that user.

F. Inventory and Notification Module

All items, whether identified through the barcode-OCR pipeline or the visual freshness pipeline, are recorded in a centralized inventory that displays the remaining shelf life of each item and its current status (fresh, nearing expiry, or expired). The system issues configurable, digest-style notifications a user-defined number of days before an item's predicted expiry, grouping multiple items into a single notification where possible to reduce alert fatigue.

IV. SYSTEM ARCHITECTURE

The system follows a layered architecture consisting of a mobile client layer, a processing layer, a data layer, and a

notification layer, summarized in Table I. The client layer handles image capture and user interaction; the processing layer performs barcode decoding, OCR extraction, and CNN-based freshness classification, favoring on-device inference to minimize latency and reduce dependence on continuous network connectivity; the data layer maintains the inventory in a cloud-synchronized document database; and the notification layer schedules and delivers expiry reminders.

Layer	Responsibility
Client	Image capture, barcode preview, inventory UI
Processing	Barcode decoding, OCR, CNN freshness inference
Data	Inventory storage, user accounts, sync
Notification	Scheduling and delivery of expiry alerts

TABLE I. SYSTEM ARCHITECTURE LAYERS AND RESPONSIBILITIES

V. IMPLEMENTATION DETAILS

The barcode and OCR pipeline is implemented using on-device machine learning APIs, which avoids the latency and cost of a per-scan cloud call and allows basic functionality to continue without an internet connection; barcode lookups against the open product database remain the only step that requires connectivity. Extracted date text is normalized using a regex-based parser that recognizes common separators and month representations before being converted into a standard internal date format.

For the freshness classification component, rather than training a convolutional network from scratch, the system uses transfer learning on a compact, mobile-friendly backbone (such as MobileNetV2), fine-tuned on a labeled dataset of fresh and spoiled produce images drawn from publicly available sources. Prior work in this space has shown that such datasets, spanning multiple fruit and vegetable types under varied lighting and background conditions, are sufficient to train classifiers that distinguish fresh from spoiled produce with high confusion-matrix accuracy [5]. The trained model is converted to a mobile-deployable format so that inference runs directly on the device. If the classifier's confidence falls below a set threshold — for instance, due to poor lighting or an unfamiliar item — the system falls back to a category-based shelf-life estimate, adjusted by the user-specified storage condition, ensuring the pipeline always returns a usable estimate rather than failing silently.

Component	Tool / Library
Mobile application framework	Flutter
Barcode scanning	ML Kit Barcode Scanning
OCR engine	ML Kit Text Recognition
Product lookup	Open Food Facts API
Freshness classification	TensorFlow/Keras (MobileNetV2 transfer learning); TensorFlow Lite
Inventory database	Firebase Firestore
Notifications	Firebase Cloud Messaging, local notifications

TABLE II. IMPLEMENTATION TOOLS AND LIBRARIES

VI. EXPECTED OUTCOMES AND DISCUSSION

As the system described in this paper is at the design and prototyping stage, a full field evaluation has not yet been conducted. Table III instead summarizes accuracy figures reported in the literature for the individual techniques the proposed system builds upon, which serve as reference benchmarks for the subsequent evaluation of the implemented prototype.

Technique	Reported Benchmark	Source
OCR-based expiry date extraction (general)	83.3% date-extraction accuracy	[1]
Date-parser-assisted OCR (ExpDate dataset)	97.74% recognition accuracy	[3]
CNN-based fruit/vegetable freshness classification	~97% confusion-matrix accuracy (fresh/rotten)	[5]

TABLE III. REFERENCE BENCHMARKS REPORTED IN RELATED LITERATURE

These figures suggest that purpose-built OCR pipelines can substantially outperform generic text recognition on printed expiry dates, and that CNN-based classifiers can reliably distinguish fresh from spoiled produce under varied conditions. Based on these benchmarks, the proposed system is expected to achieve comparable OCR accuracy for packaged items, since it adopts a similar on-device recognition approach, while the freshness classifier is expected to perform best on the limited set of produce types initially targeted, with accuracy likely declining for visually similar items (for example, certain fruit and vegetable pairs with overlapping color and texture), consistent with the misclassification patterns reported in [5]. The storage-condition-aware adjustment is a novel component not evaluated in prior work; its effectiveness will need to be assessed through user studies measuring whether adjusted estimates more closely track actual spoilage than fixed, condition-agnostic estimates. A formal evaluation of the implemented prototype, including accuracy measurements on a held-out test set and a small-scale user trial, is planned as part of ongoing work.

VII. CONCLUSION AND FUTURE WORK

This paper presented the design of a Food Expiry Detection and Reminder System that integrates barcode-based product identification, OCR-based expiry date extraction, and CNN-based visual freshness estimation within a single mobile application, with the goal of addressing both packaged and fresh food items under one unified pipeline. Unlike systems that treat fresh produce as a secondary, static-estimate feature, the proposed system performs a live visual scan of produce and adjusts its shelf-life prediction based on user-specified storage conditions, an approach not found in the reviewed literature. The system architecture favors on-device processing to reduce cost and dependence on continuous connectivity.

Future work includes completing the implementation of the freshness classification model across a broader set of produce types, conducting a formal accuracy evaluation against the benchmarks summarized in Section VI, and running a small-scale user study to assess whether the storage-condition adjustment meaningfully improves the accuracy of shelf-life predictions over static estimates. Further extensions may include recipe suggestions for items nearing expiry and integration with smart refrigerator sensors for automated storage-condition detection.

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